impl SelectionMeasure for RangeDivergence

Michael Shoemate

February 24, 2025

1 Hoare Triple

Precondition

Compiler-verified

- Associated Constant `ONE_SHOT`
  - `ONE_SHOT` is true if the measure supports one-shot top-k composition.
- Associated Type `RV`
  - `RV` must implement trait `InverseCDF`.
- Method `random_variable` *Types consistent with pseudocode.*
- Method a`privacy_map` *Types consistent with pseudocode.*

Caller-verified

- Method `random_variable`
  - `scale` is positive (*cannot be null due to `FBig` dtype*).
- Method `privacy_map`
  - `d_in` is non-null and non-negative.
  - `scale` is non-null and non-negative.

Pseudocode

```python
class RangeDivergence(SelectionMeasure):
    ONE_SHOT = True
    RV = GumbelRV

    @staticmethod
    def random_variable(shift: FBig, scale: FBig) -> GumbelRV:
        return GumbelRV(shift=shift, scale=scale)

    @staticmethod
    def privacy_map(d_in: f64, scale: f64, k: usize) -> f64:
        if d_in < 0:
            raise ValueError("input distance must be non-negative")

        if scale.is_zero():
            return f64.INFINITY

        return d_in.inf_div(scale).inf_mul(f64.inf_cast(k))
```

Postcondition

Theorem 1.1. The implementation is consistent with all associated items in the `SelectionMeasure` trait.

1. Associated Constant `ONE_SHOT`
2. Associated Type `RV`
3. Method `random_variable`
4. Method `privacy_map`

Proof of valid associated constant: `ONE_SHOT`. Since the proof of `privacy_map` only holds if k is less than two, `ONE_SHOT` is defined to be false. □

Definition 1.2. A random variable follows the Exponential distribution if it has density

$$f(x) = \frac{1}{\beta} e^{-z} \tag{1}$$

where $z = \frac{x-\mu}{\beta}$, μ is the shift (location) parameter and β is the scale parameter.

Proof of valid associated type: `RV`. The associated type `RV` is defined as `ExponentialRV`, which represents a random variable following the Exponential distribution 1.2. The compiler verifies that `ExponentialRV` implements the `InverseCDF` trait. □

Proof of valid method: `random_variable`. By the precondition on `scale` being positive, `random_variable` returns a valid instance of `ExponentialRV`. □

Proof of valid method: `privacy_map`. To be merged from: https://github.com/opendp/opendp/pull/1678/files □